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- 1) Which of the following options evaluates correctly $(f \cdot g)(x)$
a) $g(f(x))$ b) $g(x)f(x)$ c) $f(g(x))$ d) $f \cdot g \cdot x$
- 2) Which of the following functions models exponential decay.
a) $m(t) = Ae^{-rt}$ b) $m(t) = -Ae^{r-t}$ c) $m(t) = Ae^{-r+t}$ d) $m(t) = -Ae^{-rt}$
- 3) Considering $f(x) = (x-1)^2 + 1$. What would the new function be like after shifting 5 units to the left and 3 units down?
a) $f(x) = (x+5)^2 - 3$ b) $f(x) = (x+2)^2 - 4$ c) $f(x) = (x+4)^2 - 2$ d) $f(x) = (x+6)^2 - 4$
- 4) Given the function $f(x) = -2\sqrt{x^2 - 4} + 5$, which of the following represents its domain?
a) $-\infty < x \leq 0$ b) $-\infty < x \leq 4$ c) $-\infty < x < \infty$ d) $2 \leq x < \infty$

- 1) Solve the following equations. Simplify the answer as much as possible and check that all solutions satisfies the original equation. **(6 points each equation)**

a. $\log(3+x) + \log(x-9) = 0$

b. $2^{x-3} - 6^{x-2} = 0$

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- 2) For the functions $f(x) = \frac{5x}{x+2}$ and $g(x) = \frac{8}{x}$ find the indicated composition and reduce the expression as much as possible, **(6 points each operation)**

a. $(g \circ f)(x)$	b. $(f + g)(x)$

- 3) Use logarithmic properties to condense the following expression $2 \log_8 y - \frac{1}{3}(2 \log_8 x + \log_8 w)$. **(6 points.)**

Section 3: Function analysis (12 points)

- 1) Find the inverse function of $f(x) = \frac{3x-10}{2x-4}$. Reduce the expression as much as possible and proof your answer by evaluating the function composition $(f \circ f^{-1})(x) = x$. **(6 points)**

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- 2) Find the roots of the function $f(x) = \frac{x^2 - x - 2}{x + 2}$. **(6 points)**

Section 4: Modeling with math (30 points)

- 1) Hydrogen potential (pH) is a number used to describe the acidity or basicity of a chemical substance and is defined by the equation $pH = -\log(H^+)$, where H^+ measures the concentration of hydrogen ions moles per liter.
- i. Determine the pH of a coke if $H^+ = 1.3 \times 10^{-3}$. **(6 points)**
 - ii. What is the hydrogen ions concentration of the stomach acid with pH of 1.5?. **(6 points)**

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- 2) The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 12mg. Recalling that the mass decays exponentially, fulfill the following points
- i) Write the function that models the remaining mass after t years. **(6 points)**
 - ii) How much mass will remain after 400 years? **(6 points)**
 - iii) How much time will it take for the sample to be only a third of its initial value?. **(6 points)**

Section 5: Conic Sections (4 points)

- 1) Identify the conic section given the following general equation $37 - 24x + 4x^2 + 2y + y^2 = \frac{1}{4}$. **(4 points.)**